

Research Paper

Towards a sustainable energy future: Modeling Morocco's transition to renewable power with enhanced OSeMOSYS model

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ABSTRACT

Developing countries encounter numerous challenges, such as limited access to reliable electricity and a heavy reliance on imported fossil fuels. To satisfy the rising energy demand, which is essential for economic growth, these regions are shifting towards sustainable energy solutions. Solar and wind power have emerged as key and secure energy sources. This research develops an enhanced OSeMOSYS energy system model to examine long-term energy supply strategies, using Morocco as a case study. The proposed model addresses the specific needs of decision-makers in developing countries, enabling the achievement of renewable energy targets and optimal temporal resolution. A baseline and three alternative scenarios were developed, exploring both a restructuring plan to address existing gaps and a forward-looking policy targeting nearly 100 % renewable energy. These scenarios are assessed in terms of installed capacity, generation mix, investment costs, variable and fixed costs, storage capacity utilization, and carbon emissions. The results indicate that a 92 % integration rate for renewables is feasible at an additional cost of \$32 billion, thanks to new energy efficiency measures reducing demand by 15 % between 2030 and 2050 compared to baseline forecasts. Furthermore, an ambitious energy strategy by 2050 could achieve the lowest emissions rate of 0.29 Mt, paving the way for complete decarbonization. Technologically, investment in pumped-storage hydroelectric plants is the most viable backup option for a country dependent on natural gas imports. Our findings emphasize that a diversified energy mix and robust energy efficiency plans are essential for a rapid and feasible transition of the country's energy system. This paper offers an enhanced energy model to help decision-makers in developing countries set targets on the share of renewables in total installed capacity, thereby increasing their integration rate at minimal additional cost.

1. Introduction

Reducing CO₂ emissions and other Greenhouse Gases (GHG) is crucial in the effort to limit global warming, especially for developing countries, which possess significant Renewable Energy (RE) potential [1]. To harness this potential, it is imperative to formulate effective energy policies that promote investment in RE and their integration into national development strategies [2–5]. However, the shift to RE presents technical challenges, including the need for meticulous planning to balance energy production and demand [6]. Addressing these challenges necessitates the use of modeling methodologies to enhance forecasting accuracy and optimize energy configurations [7].

These energy forecast modeling frameworks can be categorized as

either bottom-up technical, top-down macroeconomic, or hybrid methodologies [7]. For this study, the bottom-up engineering approach is the most appropriate (see section 3.1). Low Emissions Analysis Platform (LEAP), MARKet ALlocation (MARKAL), The Integrated MARKAL-EFOM System (TIMES), Model for Energy Supply Strategy Alternatives and their General Environmental Impact (MESSAGE), and the Open-Source Energy Modeling System (OSeMOSYS) are the most prominent existing bottom-up modeling frameworks [8]. Among these models, our preference is for OSeMOSYS, which offers a high degree of flexibility, is accessible via an open-source license, and can adapt to the specific needs of different countries and regions. The OSeMOSYS model facilitates the seamless integration of data and scenarios, enabling the development of robust energy strategies that are tailored to the specific contexts of individual countries and regions [9].

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Nomenclature			
<i>Symbols</i>		<i>l</i>	time-slice
<i>AAD</i>	Accumulated Annual Demand	<i>s</i>	storage
<i>AF</i>	Availability factor	<i>rr</i>	Another region
<i>CF</i>	Capacity Factor	<i>Abbreviations</i>	
<i>CTA</i>	Capacity To Activity unit	BAU	Business as Usual
<i>D</i>	Demand for a fuel in a time-slice	CO₂	Carbon dioxide
<i>DA</i>	Demand for a fuel over the entire year	CSP	Concentrated Solar Power
<i>P</i>	The total Production of a fuel in a time slice	EE	Energy Efficiency
<i>PA</i>	The total Production of a fuel over the entire year	GDP	Gross Domestic Product
<i>RTA</i>	Rate of Total Activity	GNME	Nigeria-Morocco Europe Gas Pipeline
<i>SDC</i>	Storage Discounted Cost	GHG	Greenhouse Gases
<i>TCA</i>	Total Capacity Annual	GME	Maghreb Europe Pipeline
<i>TDC</i>	Total Discounted Cost	LEAP	Low Emissions Analysis Platform
<i>T</i>	Quantity of fuel exchanged between regions (Trade)	MARKAL	MARKet ALlocation
<i>TR</i>	Binary parameter defining the links between regions (Trade Route)	MESSAGE:	Model for Energy Supply Strategy Alternatives and their General Environmental Impact
<i>U</i>	Total Consumption of a fuel in time slice	NANES	New Ambitious National Energy Strategy
<i>UA</i>	Total Consumption of a fuel over the entire year	NMNES	New Moderate National Energy Strategy
<i>YS</i>	Year Split	ONEE	“Office National de l'Électricité et de l'Eau potable”
<i>Subscript</i>		OSeMOSYS	Open-Source Energy Modeling System
<i>r</i>	Region	PV	Photovoltaic
<i>f</i>	Fuel	PHS	Pumped Hydro Storage
<i>y</i>	Year	RNES	Ratified National Energy Strategy
<i>t</i>	technology	RE	Renewable Energy
		TIMES	The Integrated MARKAL-EFOM1 System
		T&D	Transmission and Distribution

In various studies, OSeMOSYS has been utilized extensively in developed countries to assess long-term supply strategies. For instance, Burandt et al. employed OSeMOSYS and GENeSYS-MOD to scrutinize the required transformation for decarbonizing China's energy system across several sectors [10]. Henke et al. developed an open-source electricity model for Europe [11]. Meanwhile, Kuling et al. focused on Canada, utilizing OSeMOSYS to analyze switchgrass-based biofuels and decarbonization pathways through a CLEWs model [12]. Groppi et al. utilized OSeMOSYS for optimal energy and water system planning on an Italian island [13].

In the context of developing countries, the versatility and applicability of OSeMOSYS have also been widely demonstrated. Palombelli et al. introduced storage losses for hydroelectricity and battery production in Atlantis, while Pappis et al. concentrated on achieving universal access to electricity in Ethiopia [14,15]. Similarly, Gardumi et al. combined OSeMOSYS with an input-output model to identify post-pandemic power system opportunities for Tunisia [16]. Dallmann et al. used OSeMOSYS to evaluate the consequences of deploying regional power systems in Egypt [17]. Tan et al. applied OSeMOSYS for techno-economic assumptions in modeling long-term energy systems in Vietnam [18]. Lastly, Gebremeskel et al. integrated OSeMOSYS with LEAP to model long-term electricity supply in Ethiopia [9]. However, the accuracy of energy modeling with OSeMOSYS, which is well suited to developed countries, is not directly transferable to developing countries. Indeed, planning and energy modeling assumptions designed for stable economic environments are often unsuitable for the volatile economies of the latter [8]. This inadequacy leads to imprecise and unreliable analyses and policy recommendations. To reflect the geographical dynamics, future timeframes, and economic variables relevant to developing countries, specific adjustments to energy models are essential. Adapting models like OSeMOSYS to the economic and social realities of developing countries is crucial [8]. Although some adaptation efforts have been made, further adjustments remain to be explored in the current literature.

The approach to energy planning and modeling varies significantly

between developed and developing countries, particularly concerning the integration of RE targets into future energy strategies [19]. In developed nations, studies based on minimum long-term RE production targets are considered reliable, given the availability of historical data, advanced projections, and robust economic and social information [20]. Conversely, applying the same methodology to developing countries is less reliable due to the lack of precise historical production data, limited future-oriented studies, and uncertainties in economic, social, and political realms [21]. Therefore, setting RE development targets as a percentage of total installed capacity, rather than fixed production targets or absolute gigawatt measures, can help mitigate the risk of over-estimating RE needs. This methodological approach increases flexibility in the phasing of RE installations, enabling energy systems to adapt more effectively to fluctuating demand and technological advances. It also enhances the efficiency and reliability of energy production from existing and planned renewable sources by adjusting to immediate and short-term requirements. Furthermore, it affords decision-makers in developing countries the time and information necessary to enhance grid management and integrate energy storage solutions to address the intermittency of RE.

The present paper focuses on Morocco, as the country's power system faces distinctive challenges that may be relevant to other developing nations. The significance of a sustainable energy supply for socioeconomic well-being is becoming increasingly apparent. Nonetheless, Morocco still heavily relies on fossil coal, and there has been limited research on the prospective evolution of its electricity supply. Recent literature has revealed several crucial and complementary analyses. Ainou et al. observed a decline in energy security due to heightened import dependency and soaring prices without proposing forward-looking solutions [22]. Similarly, Haidi et al. and Mahdavi and Vera delved into the historical progression of RE without presenting the necessary future projections for a successful energy transition [23,24]. Meanwhile, studies such as those conducted by Schinko et al. and Boulakhbar et al. detailed techno-economic scenarios and challenges related to RE integration [25,26]. However, these studies lacked well-

rounded modeling frameworks and primarily relied on expert opinion. Šimelytė highlighted socioeconomic and financial challenges but without sufficient simulations [27]. Bahetta and Hasnaoui explored the impacts of a carbon tax and emissions caps, yet overlooking crucial elements such as post-pandemic developments, strategic renewable targets, and Energy Efficiency (EE), thereby limiting their modeling [28]. While these studies are informative, they possess methodological shortcomings that limit their ability to provide comprehensive, forward-looking solutions for a sustainable energy transition. Confronting Morocco's intricate energy challenges necessitates the development of integrative and forward-looking approaches encompassing all socio-economic, technological, and environmental variables.

The primary aim of this study is to identify potential pathways and conduct a quantitative analysis of the future power generation sector in Morocco, taking into consideration the specific needs of developing countries. The goal is to provide an optimal representation of these countries' power systems, incorporating their unique characteristics, and to evaluate alternative technologies and policy choices. This paper seeks to address the following questions:

- What are the most cost-effective future supply mixes that ensure adequacy, reliability, and reduced greenhouse gas emissions?
- What will be the repercussions of delayed investment in planned RE projects, and how will the integration of these energies evolve without a recovery strategy to address the accumulated backlog?
- How will implementing an energy strategy targeting 100 % RE by 2050 impact future investments, considering the country's financial capabilities, technological readiness, environmental constraints, and overall political objectives?
- What is the impact of introducing EE policies on future energy investments?

To address these questions and fill the gaps in the existing literature, this study distinguishes itself through its innovative and integrated approach to modeling and projecting energy demand, incorporating socio-economic and technological variables specific to the Moroccan context. The key contributions of this research are as follows:

- **Policy Framework:** This research provides a comprehensive policy framework tailored to the specific needs of developing countries, addressing the gap in existing policies that often overlook the unique socio-economic and environmental conditions in these regions. By offering context-specific strategies, this framework ensures effective and sustainable RE initiatives, supporting a resilient and inclusive energy transition.
- **Advanced Planning Tools:** This study develops advanced planning tools crucial for balancing energy production and demand in developing countries. These tools enhance forecasting accuracy and optimize energy configurations, significantly improving the management of energy resources. Such advancements are essential for integrating RE sources, ensuring stability and reliability in the energy supply.
- **Adaptation of OSeMOSYS Model:** The adaptation of the OSeMOSYS model better reflects the volatile economic conditions of developing countries. Traditional models often fail to capture these unique dynamics, resulting in less reliable policy recommendations. The adapted model provides more precise and applicable insights, aiding in the formulation of effective energy policies and strategies.
- **Adjustments to Energy Models:** Introducing specific adjustments to energy models better represents the geographical, economic, and social dynamics of developing countries. Existing models frequently overlook these factors, leading to less accurate energy planning. Tailoring models to the realities of developing nations enhances the precision of energy modeling, allowing for more effective planning and decision-making.

- **Flexible Methodology for RE Targets:** This study applies a flexible methodology for setting RE development targets in developing countries, addressing the inadequacy of approaches designed for developed nations. By adopting flexible targets, this methodology allows for better adaptation to fluctuating demand and rapid technological advancements, ensuring reliable and consistent energy production.
- **Analysis of Morocco's Power System:** A comprehensive and forward-looking analysis of Morocco's power system, incorporating socio-economic, technological, and environmental variables, is provided. Existing analyses often lack this depth, limiting their usefulness for policy-making. This thorough approach informs better policy decisions by highlighting the complex interactions between various factors, supporting a more effective and sustainable energy strategy for Morocco.

The remainder of this article is divided into several sections. [Section 2](#) will provide a comprehensive overview of the country's socio-economic situation, objectives, challenges, and prospects for sustainable energy transition. It will also delve into the current state of Morocco's power system. Moving on, [Section 3](#) will outline the selection process and modeling framework used in the study, along with the data, assumptions, and scenarios employed. [Section 4](#) will then present the key findings of the modeling analysis, including power generation capacity, supply mix, investments, costs, technology mix, and CO₂ emissions. In [Section 5](#), we will examine the implications of these results and offer policy recommendations for Moroccan decision-makers. Finally, [Section 6](#) will wrap up the article by concluding our discussion and proposing avenues for future research.

2. Sustainable energy transition in Morocco: Policy, challenges, and prospects

Morocco's population is expected to grow from 35 million in 2019 to 44 million by 2050. Despite efforts to promote RE and EE, the country faces challenges with energy intensity and dependency on imported fossil fuels, impacting electricity production costs and the national balance of trade [1,29,30]. (See the [supplementary document](#) for more on Morocco's economy).

2.1. Moroccan electricity system

Morocco's electricity sector has undergone significant transformation in recent decades, thanks to a combination of policy reforms, infrastructure investment, and a focus on RE sources. [Figure S1](#), which can be found in the [supplementary document](#), provides a comprehensive overview of this power system. Through a diagrammatic representation, the figure illustrates the existing electricity infrastructure and potential future extensions, showcasing the diverse range of energy technologies and the energy carriers that connect them. The diagram's design allows for flexibility in accommodating future extensions while representing the existing energy system. By highlighting the interconnected pathways and links spanning from primary energy supply to the final delivery of demand-driven products and services through intermediate energy conversion processes, it provides a clear visual of energy flows. The subsequent paragraphs offer a comprehensive overview of the Moroccan power system, beginning with an examination of primary energies and emphasizing the potential of RE. The text also covers existing and proposed generation capacity, as well as the Transmission and Distribution (T&D) network, concluding with the current breakdown of electricity demand across different sectors.

The world's primary energy resources encompass various fossil fuels such as oil, coal, and natural gas, as well as renewable sources like nuclear power, solar, wind, and hydropower. In Morocco, these resources are categorized into six types: non-renewables, including natural gas, oil, and imported coal, and renewables such as solar, wind, and

hydropower. Despite its considerable RE resources, particularly in solar and wind, the country remains heavily reliant on imported fuels to meet its non-renewable energy demands [27,31].

The journey towards RE in Morocco began with hydropower in 1945, and this legacy continues today. The current hydropower capacity is 1800 MW, contributing between 5.1 % and 13.7 % of the country's electricity consumption, depending on rainfall. In 2016, 125 new sites for small hydropower plants were identified, with a potential capacity of 300 MW. This study focuses on the potential of small-scale hydropower plants with a capacity of up to 20 MW. The solar energy potential is substantial, with an annual exploitable capacity of 15,155 terawatt-hours (TWh) for solar PVs and 15,127 TWh for Concentrated Solar Power (CSP). The efficiency of these technologies depends on solar irradiation, yields, and conversion factors. The technical potential for both PVs and CSP is calculated using specific formulas that consider resource availability, efficiency, spacing factor, and available area (see [supplementary document-ANNEXE B](#)). Wind power also shows significant promise, with a global potential of 25 GW and an estimated annual technical potential of 1458.8 TWh, assuming a 30 % capacity factor. Calculations for wind power depend on the average annual wind speeds and their correlation with turbine load hours (see [supplementary document-ANNEXE B](#)). [Table 1](#) summarizes the country's RE potential.

Morocco utilizes various technologies for power generation, encompassing coal-fired, natural gas-fired, oil-fired, diesel-fired, and hydroelectric plants, along with renewable technologies such as solar and wind. Since 2016, there has been a significant surge in the integration of RE, reaching 37.1 % of the electricity generation mix in 2020. [Table 2](#) succinctly outlines the current electricity mix in Morocco's power system, providing details on fuel sources, installed capacity for 2019 and 2020, as well as capacity projections for 2025 and 2030. Furthermore, [Table S2](#) in [Appendix B](#) of the [supplementary document](#) presents a comprehensive inventory of operational and planned power plants in Morocco. To appraise energy storage options, two distinct modalities were considered: thermal energy storage linked to solar CSP systems and Pumped Hydroelectric energy Storage (PHS).

The power grid comprises distinct T&D systems, each serving unique roles. The transmission system transports electricity from generation points to substations and facilitates inter-regional electricity exchange through high-voltage lines. On the other hand, the distribution network connects these substations to end consumers, facilitating the final delivery of electricity to consumers. In Morocco, the power grid incorporates a mix of old and new technologies, presenting maintenance challenges due to extreme environmental conditions. The “*Office National de l'Electricité et de l'Eau (ONEE)*” is responsible for managing the network, which encompasses 27,515.9 km of high-voltage lines, 94,243 km of medium-voltage lines, and 244,514 km of low-voltage lines, with a total installed capacity of approximately 26,307 MVA. Moreover, Morocco plays a significant role in sustainable electricity trade, boasting important interconnections with Spain and ongoing projects with Portugal ([supplementary document](#)).

Electricity demand in Morocco is segmented among various consumer sectors, including industry, agriculture, services, the residential sector, and transport. In 2019, the residential sector, largely due to Morocco's status as a developing country, exhibited the highest electricity consumption at 15,664 GWh. The industrial sector followed with

Table 2

The Moroccan power system installed capacity history from 2019 to 2020 and projections for 2025–2030 [34,35].

Technologies	Capacity (MW)					
	Existing				Predicted Values	
	2019	%	2020	%	2025	2030
Coal	4116	38.6	4116	38.7	4116	4116
Natural Gas Open Cycle Gas Turbine (OCGT), Combined Cycle Gas Turbine (CCGT)	834	7.8	834	7.8	834	834
Oil-Fired	600	5.6	300	2.8	300	300
Steam Heat Plants	1110	10.4	1110	10.4	1110	1110
Diesel turbine	316	3.0	314	3.0	314	314
Hydro	1306	12.2	1306	12.3	1540	1680
Pumped Hydro Storage (PHS)	464	4.3	464	4.4	810	1110
Wind onshore	1220	11.4	1430	13.5	3560	3560
Solar PV (utility-scale)	21	0.2	61	0.6	1000	1700
Solar CSP	690	6.5	690	6.5	690	990
Total	10,677		10,627		14,272	15,719

*Note: A complete list of current and future projects in the electricity sector, together with detailed information, can be found in the supplementary documents.

a consumption of 12,700 GWh. The service and agricultural sectors took the third and fourth positions, respectively, with consumption figures of 5,839 GWh and 4,608 GWh. In comparison, electricity consumption in the transport sector remained notably low.

2.2. Morocco climate target

Morocco aims to reduce its GHG emissions by 17 % below baseline levels by 2030, in line with the Paris Agreement. With international assistance, the target is a 25 % reduction below baseline, totaling a 42 % reduction. This ambitious goal is supported by the implementation of a low-carbon development plan for 2050. The success of the national energy strategy and investor interest in green energy projects have prompted the government to raise targets for RE and EE.

The country is well-positioned to significantly increase the adoption of RE across various industries, thereby reducing dependence on oil and coal. Since the adoption of the National Energy Strategy in 2009, there has been a strong commitment to sustainable industrial development. Initially, the strategy targeted a renewable electricity share of 42 % of total capacity by 2020, later increased to 52 % by 2030. Currently, efforts are being made to exceed this objective.

3. Materials and methods

This section delineates the methodology employed to analyze the prospective configuration of the Moroccan power system incorporating RE facilities. It establishes the foundation for comprehension of the modeling approach and the study's underlying assumptions. Subsection 3.1 elucidates the selected modeling framework, followed by section 3.2, which presents the model's current structure and proposed enhancements. Section 3.3 examines various potential pathways for Morocco's energy transition. In addition, section 3.4 explains the input data used for the model and parameter calibration, while section 3.5 discusses model validation and verification. A visual summary of the methodology is provided in [Fig. 1](#).

3.1. Choice of modeling framework

The selection of an appropriate modeling framework necessitates an evaluation of contextual factors, challenges, and policy issues [36]. In the realm of energy system modeling, three main strategies prevail: bottom-up technical, top-down macroeconomic, and hybrid

Table 1
Moroccan Renewable Energy potential [1,25,32,33].

Technologies	Unit	Exploitable reserve
Hydropower (Small)	MW	54
Solar Photovoltaic (PV)	Twh/yr	15 155
Concentrated Solar Power (CSP)	Twh/yr	15 127
Wind	Twh/yr	1458.8 (CF 30 %/yr)11 297 (CF 40 %/yr)851 (CF 20 %)

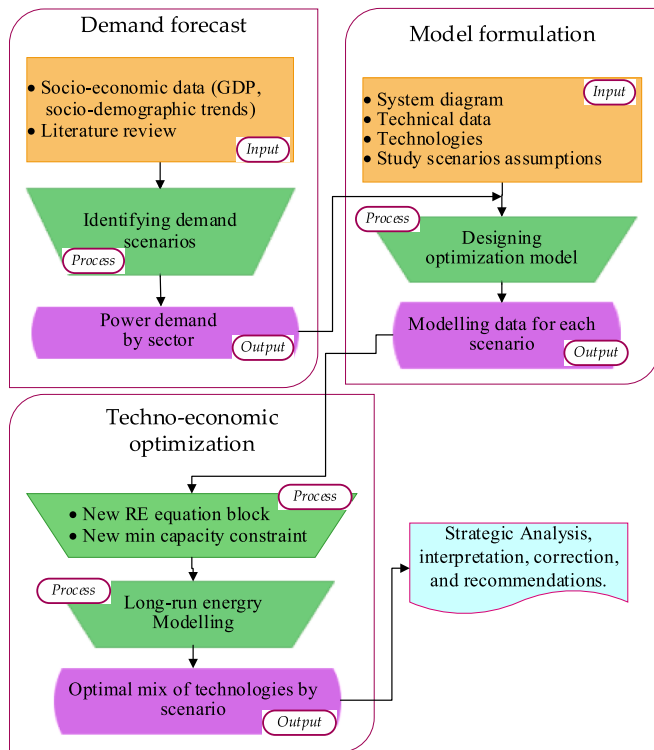


Fig. 1. Methodology used to conduct the study and develop policies for the new energy strategy.

methodologies. Top-down models analyze macroeconomic policies and energy price impacts on the global economy, but often lack detailed descriptions of energy technologies and must incorporate ongoing technological advances [37,38]. Conversely, bottom-up models offer a significant advantage for optimal planning. They provide precise technical details, enabling cost-effective planning for balancing energy supply and demand [39]. Although hybrid models incorporate elements of equilibrium and disequilibrium to address short- and medium-term market imbalances, this falls outside the scope of our current study. Therefore, to assess the impact of different energy strategies on technological choices for power generation in a developing country like Morocco, the bottom-up engineering approach proves most fitting. This method stands out for its flexibility and ability to accommodate disparities between traditional and modern energy sources [8].

Numerous tools are available to support energy system analysis through the use of bottom-up optimization models. Some of these tools include MESAP, MESSAGE, TIMES, MARKAL, and OSeMOSYS [8]. Each model comes with its own distinct features, and the selection of a particular model depends on its compatibility with specific application requirements. Among these, TIMES, MESSAGE, and OSeMOSYS are widely utilized on a global scale.

When evaluating these models, it is important to consider factors that are relevant to developing countries, such as energy consumption patterns, economic development, electrification rates, infrastructure deficiencies, supply shortfalls, data availability, and required expertise [40,41]. MESSAGE and OSeMOSYS share several similarities, including an investment focus, a bottom-up methodology, a long-term horizon, scenario analysis, and a consideration of traditional energy sources [41]. Nevertheless, OSeMOSYS's open-source nature confers a strategic advantage, enabling adaptability and customization to specific user needs. OSeMOSYS has demonstrated efficacy in long-term optimization in several countries, including Egypt, Tunisia, Ghana, Saudi Arabia, Iran, and Bangladesh [16,17,42–45].

In light of these considerations, we selected OSeMOSYS to assess the dynamics of Morocco's electricity supply from 2020 to 2050.

OSeMOSYS, designed to operate under the assumptions of perfect foresight and competition, is well suited to the operational context of Morocco's electricity sector, which is managed by a single public operator [46]. This makes OSeMOSYS a suitable choice for long-term energy planning and optimization in Morocco.

We have developed an advanced version of OSeMOSYS that takes into account the electricity infrastructure of Morocco, incorporating new RE constraints and modifying existing ones. Four scenarios have been created to consider policy decisions, variations in power generation technology costs, potential power plant commissioning schedules, and the country's energy sovereignty goals. This approach, informed by extensive research and consultation with Moroccan energy sector stakeholders, considers several factors, including domestic energy resources, the technical attributes of wind and solar technologies, electricity supply infrastructure, end-user demand patterns, and the techno-economic characteristics of the technologies involved.

3.2. Current model structure and enhancements

The model constructed in this study is designated as OSeMOSYS-Morocco. It was developed using the OSeMOSYS long-term energy system optimization tool to meet the specific needs of the Moroccan energy system. OSeMOSYS is an advanced linear optimization tool that utilizes a linear program-based algorithm to identify the most cost-effective combination of power generation technologies and capacities for investment and energy dispatch. It takes into account various policy scenarios and integrates different energy-consuming sectors to minimize costs over a long-term period.

The OSeMOSYS model is methodically structured into seven functional blocks: objective specification, cost considerations, storage parameters, capacity adequacy, energy balance, constraints, and GHG emissions. Each block includes specific equations and inequalities that can be selectively incorporated based on the modeled case to represent different parts of an energy system (Figure S2 in the supplementary document). The flexible structure of OSeMOSYS allows energy system analysts to establish unique, country-specific energy objectives. Furthermore, its open-source capabilities have been utilized by numerous researchers to enhance its features and adapt it to meet their analytical requirements [36,47]. In addition, functional modules have been released to improve the model's performance [48,49].

A pivotal element of the OSeMOSYS model is the “renewable activity” block, which is designed to ensure that RE production aligns with specific user-defined targets. However, establishing these targets presents notable challenges [19]. It is incumbent upon decision-makers to establish a minimum ratio of RE production to total electricity production. This task is complicated by the variability of capacity factors for renewable technologies due to climatic conditions [20]. Moreover, precise target setting is hindered by uncertainties in demand projections and the availability of reliable historical data. This challenge is particularly evident in developing countries, where data on past production is often limited, and prospective studies are limited [21]. At the same time, uncertainties surrounding economic, social, and political developments are high, leading to less reliable study outcomes. Thus, obtaining comprehensive data and utilizing advanced projections are crucial for accurate and dependable energy system modeling in these regions.

In this study, we introduce a novel model incorporating RE development targets as a percentage of total installed capacity. This approach is designed to mitigate the risk of overestimating RE needs in developing countries. We developed a new functional sub-block, “Renewable capacity” (refer to Figure S2 in the supplementary document), to enhance the flexibility of RE installations. This approach allows energy systems to better adapt to demand fluctuations and technological advances, improving the efficiency and reliability of energy production from both existing and planned renewable sources by meeting immediate and short-term needs. Moreover, it affords decision-makers in developing countries with the necessary time and information to enhance grid

management and integrate energy storage solutions, addressing the intermittency challenges of RE sources.

The OSeMOSYS model is founded on a complex system of equations that harmoniously incorporate diverse elements of energy generation, distribution, and consumption. It utilizes a least-cost optimization method to calculate investment and dispatch strategies that result in the lowest overall costs for the energy system. The model incorporates various constraints, such as capacity and availability, energy generation and resource availability, and emission management through penalties and limits. Additionally, the model considers a discount rate and salvage value for investments, calculating the discounted total investment costs for all technologies. By minimizing the total system cost while respecting these constraints, OSeMOSYS ensures a cost-effective and efficient energy system configuration. The central model of OSeMOSYS is based on the following system of equations:

$$\min \sum_{y,t,r} \left(\sum_t TDC_{r,t,y} + \sum_s SDC_{r,s,y} \right) \quad (1)$$

Subject to:

$$RTA_{r,t,l,y} \leq TCA_{r,t,y} \times CF_{r,t,l,y} \times CTA_{r,t} \quad (2)$$

$$\sum_l RTA_{r,t,l,y} \times YS_{l,y} \leq \sum_l TCA_{r,t,y} \times CF_{r,t,l,y} \times YS_{l,y} \times AF_{r,f,y} \times CTA_{r,t} \quad (3)$$

$$P_{r,l,f,y} \geq D_{r,l,f,y} + U_{r,l,f,y} + \sum_{rr} T_{r,rr,l,f,y} \times TR_{r,rr,f,y} \quad (4)$$

$$PA_{r,f,y} \geq UA_{r,f,y} + \sum_{rr} T_{r,rr,l,f,y} \times TR_{r,rr,f,y} + AAD_{r,f,y} \quad (5)$$

The cost component in equation (1) is divided into technology costs $TDC_{r,t,y}$ and storage capacity costs $SDC_{r,s,y}$, depending on specific sets such as technology (t), geographic region (r), and year (y). Constraints (2) to (5) are essential for maintaining system equilibrium, where (f) represents a fuel or commodity used by another technology in the model. The primary operational constraints, namely capacity adequacy (Equation (2)) and energy balance (Equation (4)), are of paramount importance. Equation (2) ensures that models invest in the necessary capacity for all technologies. Equation (3) extends this adequacy to the annual level. Equation (4) ensures that supply equals or exceeds demand at every time step, while equation (5) maintains this balance at the annual level. Please refer to the [supplementary document](#) and the nomenclature provided above for a more detailed understanding of each term.

Box 1 shows the equations for the new “Renewable capacity” sub-block. In our analysis, equation (S6) distinguishes the total capacity of renewable technologies from other capacities in the power system. Equation (S7) separates power generation from other technologies. Furthermore, the new constraint introduced by inequality (S8) enables users to set the share of installed capacity of RE sources in energy policy. This enhances the model’s adaptability and precision in integrating RE. A detailed explanation of the method used to develop these equations is provided in the [supplementary document](#).

Furthermore, another constraint in OSeMOSYS has been enhanced,

namely the minimum capacity of a given technology. Originally, this constraint allowed analysts to define a fixed minimum capacity in GW to influence the model towards certain technologies considered significant. However, accurately projecting long-term minimum capacity requirements can be challenging for decision-makers in developing countries. They often prefer to define a proportional share of each technology in the total installed capacity. In our assessment of the Moroccan case, we encountered this challenge when trying to define the minimum share of 20 % for solar PV and wind technologies and 12 % for hydropower required to achieve the target of 52 % of total installed capacity by 2030. To address this issue, we have expanded the modeling capabilities of OSeMOSYS-Morocco with an additional constraint, presented in equation (S10). This constraint allows analysts to explore different scenarios and determine the minimum required share of technology in the total installed capacity. Further details regarding this constraint can be found in the [supplementary document](#).

OSeMOSYS has been formulated in several programming languages, with GNU MathProg playing a significant role in its initial development. Over time, the code base has evolved to incorporate GAMS and Python [50]. For this study, we employed the Python version of OSeMOSYS. Specifically, we employed Python with the Pyomo framework to introduce enhancements and run simulations. To aid interested readers in understanding and reproducing this implementation, we have included code snippets in Figure S3 of the [supplementary document](#).

3.3. Scenario's design and description

In this section, we explain the scenarios analyzed and how they were developed.

Effective planning at both national and global levels is crucial for sustainable energy system development. This requires creating various scenarios that consider three significant uncertainties: geopolitical and economic developments, accessibility and technological advances, and societal attitudes and behaviors [51]. These uncertainties have a direct impact on multiple factors, such as price trends, innovation, demand growth, financing, political strategy, and more [52]. To learn more about these uncertainties and their implications for the Moroccan energy system, please refer to the [supplementary document](#). As part of our research, we analyzed reports from national and international organizations and interviewed local stakeholders to gather their opinions and perspectives. We delved into the outcomes of the National Energy Strategy (NES) introduced in 2009 and sought guidance from experts in the field to identify the key factors contributing to its success [1,22,31]. Our ultimate goal was to determine if a new strategy was necessary for 2050 or if tweaking the current strategy sufficed to achieve the 2030 objectives.

In this section, we will explore four potential futures for the Moroccan electrical system up until 2050. These futures are categorized into four scenarios: Business As Usual (BAU), Ratified National Energy Strategy (RNES), New Moderate National Energy Strategy (NMNES), and New Ambitious National Energy Strategy (NANES). Each scenario is heavily influenced by a variety of factors, including the achievement of RE integration goals, implemented policies, the availability of financing

Box 1

Equations for the new block developed.

$$TReCA_{r,y} = \sum_t (TCA_{r,t,y} \times ReTgT_{r,t,y}) \quad (S6)$$

$$TPCA_{r,y} = \sum_t (PTT_{r,t,y} \times TCA_{r,t,y}) \quad (S7)$$

$$RemCT_{r,y} \times TPCA_{r,y} \leq TReCA_{r,y} \quad (S8)$$

$$TCA_{y,t,r} \geq (TAmC_{y,t,r} + (TAmCS_{y,t,r} \times TPCA_{y,r})) \times PTT_{y,t,r} \quad (S10)$$

for future RE projects, the level of EE achieved, and the evolution of technological costs over time. The BAU scenario assumes that current RE policies will remain unchanged, with possible delays in the implementation of solar, wind, and hydroelectric projects. It also assumes that old coal-fired power plants will not be replaced and that the development of new gas-fired power plants will be suspended until 2040. This scenario does not take into account any institutional or electricity sector reforms that could accelerate growth, nor does it consider any new energy policy measures, such as RE targets or regulations on Carbon dioxide (CO₂) emissions from the electricity sector. This scenario is provided as a benchmark for comparison with other alternative scenarios to assess their impact.

The RNES scenario outlines a plan to revitalize the energy sector and achieve specific objectives by 2030. The plan involves sourcing over 52 % of the country's total installed capacity from RE sources, including 20 % from solar, 20 % from wind, and 12 % from hydro. To ensure these targets are met, significant support will be provided to RE projects and supporting infrastructure, with strict adherence to set deadlines to avoid any delays. Fossil fuel power plants will continue to operate until their planned end of life, while coal-fired power plants will not be renewed. Additionally, the development of new gas-fired power plants will be postponed until 2040. From 2030 onwards, the energy model will evolve without technological limitations regarding RE. It is important to note that there are no limitations on CO₂ emissions. Finally, the RNES scenario aims to reduce annual electricity consumption by 5 % through improved EE compared to baseline projections (Figure S10 in the supplementary document).

The NMNES scenario aims to significantly increase the use of RE and implement stronger environmental protections by 2050. By 2040, the plan anticipates that 57 % of all installed capacity will come from renewable sources, with an expected increase to 67 % by 2050. To achieve these goals, there will be a gradual reduction in CO₂ emissions beginning in 2025, as well as the development of natural gas power plants to complement RE and promote energy conservation. The NMNES also includes a voluntary 10 % reduction in annual electricity consumption through EE measures, as outlined in Figure S10 of the supplementary document.

The NANES scenario takes a step further than the NMNES scenario by presenting a more positive outlook for achieving RE goals. The focus of NANES is to increase renewable capacity while keeping costs low, to achieve a contribution from RE of at least 72 % by 2040 and 92 % by 2050. Additionally, a progressive tax on CO₂ emissions will be implemented from 2025 onwards to reduce the country's carbon footprint (as

shown in Table S5 of the supplementary document). Lastly, NANES sets a strategic objective of reducing annual electricity consumption by 15 % compared to the baseline projection (as seen in Figure S10 of the supplementary document). For an overview of the scenarios and their key parameters, refer to Table 3 below.

The four scenarios share a common set of assumptions, which include utilizing a PHS system, a solar CSP storage system, and gas-fired power plants to maintain a reliable supply of RE. Beginning in 2021, power plants will be fueled by natural gas imported from Spain. By 2025, local sources (LOCAL) will be introduced, providing roughly 300 mcm/year over a ten-year period. By the year 2035, the system will be capable of integrating Nigerian gas via the Nigeria-Morocco Europe Gas Pipeline (GNME) pipeline. Additional information regarding these assumptions can be found in the supplementary document.

3.4. Data and main assumptions

In this section, a comprehensive overview of the data inputs used in the model is provided, highlighting the significance of precise data collection for the model's dependability. A diverse array of data was gathered, encompassing socio-economic data (population growth, Gross Domestic Product [GDP] projections, and urbanization trends), energy data (current consumption, production capacities, and future demand forecasts), technology data (costs, efficiency rates, and emission factors), and policy and regulatory data (existing policies, regulations, and targets). Given the absence of a systematic energy data collection system in Morocco, the data collection process drew from both Moroccan and international sources. Consequently, the study relied on information from public reports on the Moroccan power sector [24,26,35], as well as international studies and reports [1,33,53–57]. To ensure accuracy and reliability, fundamental assumptions were established, including the use of USD 2019 as the currency unit, sinking fund depreciation, and a 5 % discount rate applied consistently across all technologies. Table 4 provides the data and assumptions used in this study. Additionally, the study assumes that no new coal-fired power plants will be built, and all existing coal-fired power plants will be decommissioned upon reaching the end of their current service lives.

In the following paragraphs, we discuss in detail each of the hypotheses presented in Table 4.

Demand projections: As a secondary objective, our research develops an electricity demand projection by analyzing socio-economic data and potential adaptations to future changes rather than relying solely on linear regression of historical data.

Table 3
Overview of scenario analysis and differences in key parameters.

Scenarios Parameters	Business As Usual	Ratified National Energy Strategy	New Moderate National Energy Strategy	New Ambitious National Energy Strategy
Evolution of the Annual Electricity Demand	In line with baseline projections (section 3.4)	5 % reduction on baseline projections	10 % reduction on baseline projections	15 % reduction on baseline projections
Future commitments and public policies.	No national policy reforms	Actions to ensure timely completion of scheduled projects.	- A new energy strategy with moderate targets for 2050. New EE policies	- A new energy strategy with ambitious targets for 2050. New EE policies
Delays in scheduled projects	– 3 years for solar – 2 years for wind power	No delays	No delay	No delay
Storage capacity and back-up	• PHS • Solar CSP • Gas PP			
Natural gas supply	• Maghreb Europe Pipeline (GME) (from Spain) • Local field (300 mcm/year for ten years) (available in 2025) • Nigeria Morocco Europe Gas Pipeline GNME (from Nigeria) (available in 2035)			
Development of new gas-fired power plants	From 2040	From 2040	From 2030 onwards	From 2030
CO₂ emissions	No limitations	No limitation	- Limitation of CO ₂ emissions	- A progressive penalty from 2025
Renewable Energy constraints	No constraints	• 52 % of installed capacity in 2030 • 20 % solar, 20 % wind and 12 % hydro. • No constraints after 2030	• 57 % of installed capacity in 2040 • 67 % in 2050	• 72 % of installed capacity in 2040 • 92 % of installed capacity in 2050.

Table 4
Main OSeMOSYS-MOROCCO assumptions.

Data	Assumptions
Modeling horizon	The proposed model was employed to identify economically viable transitions for the Moroccan National Electrical System (MNES) spanning from 2020 to 2050. In order to prevent any potential “side effects” from restricting the model’s scope and precision, the modeling was expanded to cover the duration from 2020 to 2055 with yearly accounting intervals. This guarantees a thorough and impartial methodology, as opposed to a narrow-minded “end-of-time” viewpoint.
Demand Projections	We have conducted an analysis of socio-economic data to formulate three demand forecast scenarios: a low-growth scenario, a high-growth scenario, and a reference scenario. As a result, our projections indicate an annual increase in electricity demand of 5.3 % by 2023, 3.9 % by 2030, and 2.6 % by 2050. Anticipated demand surpasses 46 TWh in 2030 and 77 TWh in 2050. Peak demand is expected to reach 10 GW in 2030 and 16.5 GW in 2050. The model’s electricity demand profile is estimated using the 2019 load curve, which is then modified to account for seasonal variations and different times of the day [35].
Temporal resolution	The demand for electricity is analyzed using a comprehensive time frame that encompasses four seasons, two types of days, and eight-time slots. This approach provides greater temporal resolution for optimization. (See supplementary document for more details.)
Fuel prices	The estimated fuel prices employed in the model are presented in the supplementary document (Table S9).
Power generation capacity	Table 2 summarizes the current electricity mix by fuel source, installed capacities for 2019 and 2020, and projected capacities for 2025 and 2030, while the model includes only contractually committed or under-construction projects, with technical and economic parameters detailed in Table S7 of the supplementary document.
Power grid	The Moroccan power grid faces challenges like extreme weather and construction damage, prompting ONEE to improve infrastructure. Despite initial significant power losses, these decline over time, with the unified system’s T&D costs detailed in supplementary tables.
Technology Capacity Factors (CF)	• <i>Conventional technologies</i> • Private concessionaires have “take-or-pay” contracts with ONEE, which gives an estimated CF of 85 % for coal-fired power plants. Natural gas-fired plants have a lower average CF of 80 %, which is expected to decrease due to geopolitical disruptions, leading to an estimated CF of 75 %. [1,33,35]. • <i>Renewable technologies</i> • After conducting thorough research on PV panels and wind turbines in multiple locations across Morocco, we have been able to estimate the daily CF of renewables in our model. Consequently, we have integrated diverse CF profiles for renewables throughout the year, which can be found in Table S8 of the supplementary document. Nevertheless, we have not factored in the variability of hourly RE production profiles between different years. • For further details on the establishment of the CFs, please refer to the supplementary document.
Emissions Factors	Morocco is committed to tackling climate change, aiming to reduce GHG emissions by 42 % by 2030, with specific targets for the power sector, and continuing this positive trend until 2050.

According to a recent study by the World Bank, accurate load forecasts depend on experience and the use of optimal methodologies [58]. While econometric projections are typically more precise than heuristic models, certain countries may encounter obstacles such as rapid economic changes, conflicts, or natural disasters. Therefore, it is essential to consider each country’s unique circumstances to obtain reliable results and utilize multiple forecasts of future load.

In our case study, we examined socio-economic data, including GDP and socio-demographic trends, to establish three demand forecast scenarios: a low-growth scenario, a high-growth scenario, and a reference scenario. The reference scenario is based on an average annual GDP growth rate of 3 % over the modeling period, in line with forecasts by reputable organizations such as the Haut-Commissariat au Plan (HCP), the World Bank, and the International Monetary Fund [29,30]. These projections indicate that the population will increase from 35.6 million in 2019 to 43.6 million in 2050, with the number of households rising from 8.3 million to 13.7 million during the same period. Additionally, urbanization is expected to increase from 62.9 % to 73.6 % by 2050.

The high-growth scenario assumes comparable demographic trends to the reference scenario but with a 4 % higher economic expansion over the modeling period. The low-growth scenario also supposes similar demographic trends but mediocre economic expansion, resulting in a GDP growth rate of 2.5 % over the same period. We incorporated this data into the International Atomic Energy Agency’s MAED-2 model [59], which allowed us to formulate three electricity demand scenarios:

- A baseline scenario with an average annual load growth of 3.9 %;
- A high-growth scenario with an average annual load growth of 5.3 %;
- A low-growth scenario with an average annual load growth of 2.6 %;

This methodology has been extended to 2050, with three distinct load curve shapes based on low, medium, and high GDP projections (refer to figure S8 in the supplementary document). We have divided this annual demand into 64-time slices to capture seasonal and daily fluctuations while maintaining temporal resolution (refer to Figure S9 in the supplementary document). Table 5 provides a summarized view of annual electricity demand in the various sectors throughout the modeling period. (See the supplementary document for more details on how we project electricity demand.)

Temporal resolution: In order to account for fluctuations in demand and production, the year is segmented into seasonal, weekly, and daily intervals. Each season is comprised of two typical days (weekdays and weekends). Additionally, each 24-hour day is further segmented into eight three-hour time slots, resulting in 64 time steps. This comprehensive temporal resolution corroborated by prior research [60,61], effectively captures fluctuations in demand and RE. For further details, please refer to the supplementary document.

The power generation capacities for Morocco’s electricity mix were gathered from local data sources [34,35,62], as well as the “World Electric Power Plants Database” [63]. Table 2 provides a summary of the current energy mix, including details on fuel sources, installed capacities for 2019 and 2020, and projections for 2025 and 2030. Our model incorporates only future projects that have been contractually committed or are under construction, encompassing power plants and commercial power links (Table S2 in the supplementary document). Other projects are regarded as investment options. Each power generation technology has been assigned technical and economic parameters, such as lifetime, construction time, efficiency, load factor, investment cost, operating cost, and maintenance cost. These parameters are outlined in Table S7 of the supplementary document.

The Moroccan power grid operates as a unified system across the kingdom, with electricity generation managed by a single operator. The grid is facing challenges from extreme weather and construction damage, leading ONEE to place emphasis on upgrading its infrastructure. Our analysis accounted for country-specific losses, which, while initially significant, decline consistently over time (Table S5 of the supplementary document). T&D costs are detailed in Table S6.

The Capacity Factor (CF) is a metric used to assess the efficiency of power generation assets, with values varying by plant type, namely conventional or renewable. Conventional power plants, such as those fueled by coal or natural gas, typically exhibit a controllable capacity factor (CF) that is estimated to be 85 % for coal-fired plants and 80 % for

Table 5

A summary of annual electricity demand by sector over the modeling period.

Sectors	2019	2020	2025	2030	2035	2040	2045	2050
Industry (GWh)	12 700	9 720	12 079	13 905	15 182	16 697	18 465	20 507
Service sector (GWh)	5 839	4 195	5 858	7 186	8 250	9 452	10 813	12 356
Administrative (GWh)	2 964	2 313	2 960	3 925	4 760	5 678	6 702	7 850
Residential (GWh)	12 742	10 486	12 810	15 793	18 201	20 914	23 981	27 456
Agriculture (GWh)	4 608	3 570	4 410	5 344	6 080	6 917	7 867	8 948

natural gas-fired plants. However, geopolitical tensions with Algeria may reduce the CF of gas-fired plants to 75 %. RE sources, influenced by weather, require a comprehensive approach to determine CF. We analyzed wind turbines and PV panel CFs using Moroccan and international data [27,53,57]. By assessing wind speeds and solar power plant data, we established long-term CF projections reflecting RE variability. For the 64-time slices in our study, we consulted additional sources [64,65] ensuring an accurate representation of CF variability in RE sources, as detailed in Table S8 of the [supplementary document](#).

Emissions Factors: Morocco has made notable progress in addressing climate change, as evidenced by its hosting of the COP22 conference and its support for the global phase-out of coal-fired power generation. The country aims to reduce GHG emissions by 42 % by 2030, with an unconditional target of 17 % and a conditional additional 25 % reduction contingent on international support. In 2019, the power sector was responsible for 45 % of CO₂ emissions, and Morocco plans to reduce cumulative emissions by 10.5 % between 2020 and 2030, totaling 496.7 MtCO₂e. Our research indicates that this positive trend will continue until 2050, supported by comprehensive data on fuel-specific CO₂ emission factors provided in the [supplementary document](#).

Data calibration: The model underwent calibration for the period 2015–2020 by utilizing available historical data and employing a two-stage approach. Initially, we constrained the installed capacities and production of each technology, refining the model into a quasi-simulation model to validate the inputs. The primary energy use and system costs were not constrained, resulting in an error of 0.8 % for primary energy use and 1.2 % for total discounted cost. Subsequently, by constraining only installed capacity and leaving generation free, the model optimized power generation, demonstrating a maximum error rate of 12 % for generation and costs per technology. While there is limited information in the literature regarding an acceptable error for optimization models, Dhakouani et al. discovered closely aligned error values during the calibration of the Tunisian OSeMOSYS model under constraints similar to those of the point [36]. Furthermore, the calculated errors were deemed acceptable by the Moroccan experts and stakeholders whom we consulted.

4. Results analysis

The following section presents the principal findings of the modeling analysis in relation to electricity generation capacity, the composition of the electricity supply mix, investments, operation and maintenance costs, the overall technology mix, and CO₂ emissions.

4.1. The Business As Usual (BAU) scenario

• Installed capacity

The Moroccan government has developed an energy strategy to ensure a consistent supply of electricity, which involves expanding the range of energy sources. Projections for the business-as-usual scenario predict a gradual increase in installed power generation capacity, from 10.63 GW in 2020 to 23.17 GW in 2030, 32.21 GW in 2040, and 50.23 GW in 2050 (as depicted in Fig. 2). RE are expected to make up 33 % of the energy mix in 2020, rising to 57 % in 2030, surpassing the goal of 52 %. This figure is projected to increase to 70 % in 2040 and 73 % in 2050. Wind power will play a significant role in this diversification and is expected to expand considerably, contributing 26.52 GW to total capacity by 2050. Solar PV technology will also make strides, reaching 8.62 GW by mid-century. However, solar CSP technology is anticipated decreasing and gradually phase out by 2050. Hydropower is expected to maintain a stable capacity throughout the forecast period, while the decommissioning of coal-fired power plants will make wind technology the primary source of electricity generation. To sustain the expanding use of variable renewables, it will be necessary to invest in supplementary technologies like PHS and gas-fired power plants. Over time, the capacities of PHS technologies and gas-fired power plants will increase, with projections reaching 4.35 GW and 1.11 GW, respectively, by 2040 and 9 GW and 3.7 GW by 2050. These technologies will play a critical role in ensuring sufficient generation capacity and reserve margins, especially since no advancements are anticipated for gas-fired power plants until after 2040. Diesel and fuel oil technologies will not have any new capacity added during the modeling period, but their residual capacity will be retained until they are decommissioned. Import

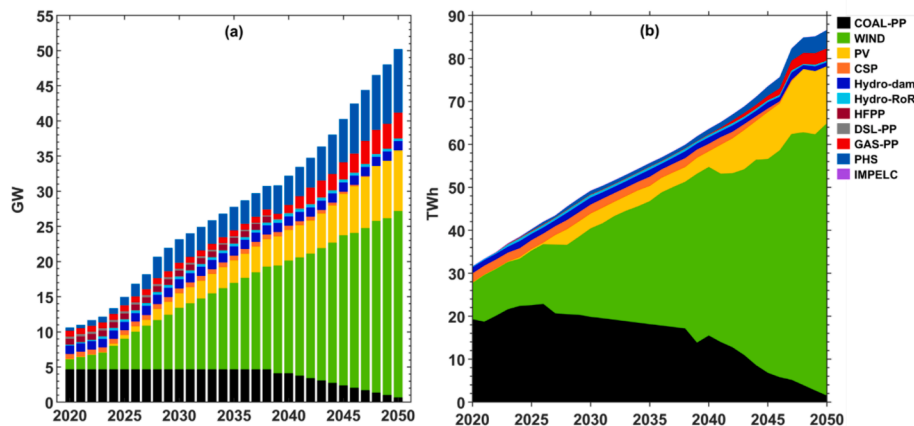


Fig. 2. Power generation capacity (a) and Electricity supply mix (b) in the Business-as-Usual scenario.

capacity will remain constant throughout the modeling period.

• Composition of Power Generation

According to the BAU projection, we can anticipate a significant increase in electricity supply over the next few decades. From 31.66 TWh in 2020, it is expected to rise to 49.27 TWh in 2030 and then gradually climb to 63.65 TWh in 2040, ultimately reaching 86.36 TWh in 2050. RE technologies will play a vital role in this growth, contributing 39 % (12 TWh) in 2020 and steadily increasing to 58 % (28 TWh) in 2030, 73 % (47 TWh) in 2040, and an impressive 90 % (85 TWh) in 2050. Although wind power will continue to be the primary source of electricity generation, accounting for 81 % (63 TWh) of the RE share in 2050, it will be followed by solar PVs with 17 % (13 TWh) and hydroelectric power with 2 % (1.28 TWh). As depicted in Fig. 2, fossil fuels will contribute only 5 % of the total electricity supply in 2050, with natural gas (3 %, or 3 TWh) and coal (2 %, or 1.5 TWh) comprising the majority of this figure. PHS will, therefore, represent the remaining 5 % (4.26 TWh) of the electricity mix in the same year.

It is important to note that as we move away from fossil fuel-based electricity generation, renewable technologies like wind power and solar PVs will gradually become more integrated into the electricity supply mix. However, in order to maintain the stability of the power system, it will be necessary to create a new PHS capacity. That is where hydraulic storage comes in; it will play an increasingly crucial role in meeting the energy demands. Specifically, we can expect load energy to grow from a modest 2.2 GWh in 2020 to a substantial 285 TWh in 2050, while discharge energy will also increase from 2.1 GWh in 2020 to 273 TWh in 2050, as illustrated in Fig. 3.

• Overall system costs

Based on the BAU scenario, capital expenditure will amount to a total of USD 137 billion between the years 2020 and 2050 (as seen in Fig. 4). The focus of these investments will center around enhancing power generation and grid infrastructure, as well as expanding hydropower storage capacity. Out of the total, USD 49 billion will be dedicated to power generation technologies, with a particular emphasis on RE, which will receive USD 43 billion. Wind power will receive USD 34 billion, solar PVs USD 7 billion, and hydroelectric plants USD 2 billion. Auxiliary technologies will be allotted USD 7 billion, including USD 4 billion for PHS (USD 1 billion for storage capacity) and USD 3 billion for gas-fired power plants.

Additionally, the extension of the power grid is expected to cost USD 87 billion over the same period. Operating and maintenance expenses

over the course of 30 years are estimated to be USD 37 billion, primarily consisting of fixed costs (USD 28 billion) and variable costs (USD 9 billion). Consequently, the total discounted cost for the modeled system will be USD 117 billion, spread out among the technologies and modeling period, as shown in Fig. 4.

4.2. Scenarios comparison

Except for the baseline scenario, it is predicted that EE measures will decrease electricity demand from 2030 onwards. Nevertheless, RE will remain a crucial component of the energy portfolio until 2050. PHS technology and gas-fired power plants will remain stable in the BAU, RNES, and NMNES scenarios. Conversely, the NMNES scenario will exhibit lower renewable capacity and different energy investment trends compared to the BAU scenario. Additionally, the NANES scenario proposes EE measures resulting in a 15 % reduction in electricity demand from 2030 onwards. However, it also foresees an increase in electricity generation capacity and RE investment by 2050 compared to the BAU scenario.

The scenarios of RNES, NMNES, and NANES exhibit diverse changes in electricity production and consumption when compared to the BAU scenario. However, they all show a similar electricity supply composition with a predominance of RE. Wind power is predicted to become the primary source of electricity by 2030, supplying 70–73 % of the total by 2050. Additionally, according to the NMNES scenario, renewable sources are expected to account for 91 % of the 89 TWh increase in electricity supply by 2050 compared to the BAU scenario. The NANES scenario forecasts a decrease in electricity supply of 2.1 TWh compared to the BAU scenario, reaching 84.26 TWh by 2050. There has also been a noticeable decline in hydropower storage usage between 2020 and 2050, which can be attributed to the development of hydroelectric projects.

The RNES scenario requires USD 3.8 billion less capital expenditure than the BAU scenario between 2020 and 2050 thanks to EE measures. Despite both scenarios having the same RE penetration rate of 52 % in 2030, the RNES scenario requires less installed capacity to meet demand. Moreover, the implementation of EE measures in the NMNES scenario will reduce demand by 10 % by 2030, saving USD 9 billion in investment costs compared to the BAU scenario between 2020 and 2050. Lastly, The NANES scenario, aiming for ambitious energy transition targets, requires up to USD 218 billion in investment. Notably, it only needs an additional USD 27.7 billion over the BAU scenario between 2020 and 2050.

The Moroccan government is committed to creating a sustainable electricity system by reducing GHG emissions from 2020 to 2050. Based

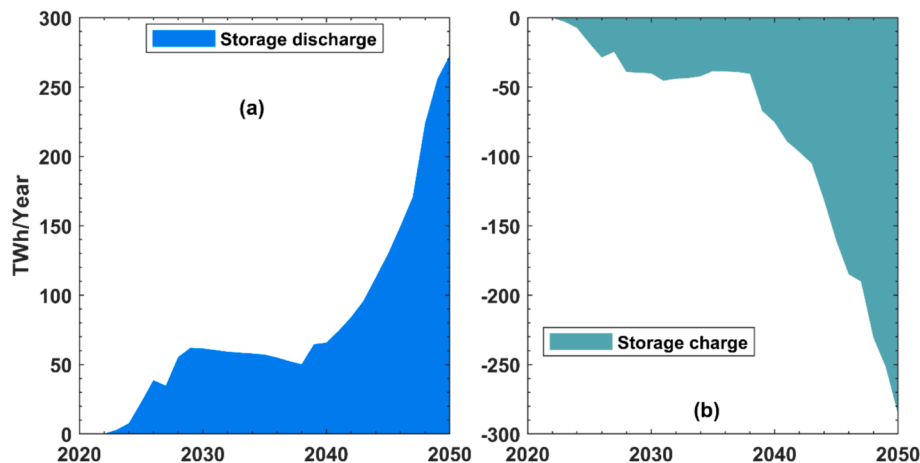


Fig. 3. Energy supplied by hydroelectric storage (a) and energy consumed to load hydroelectric generation (b) per year, Charging and discharging energy in the BAU scenario.

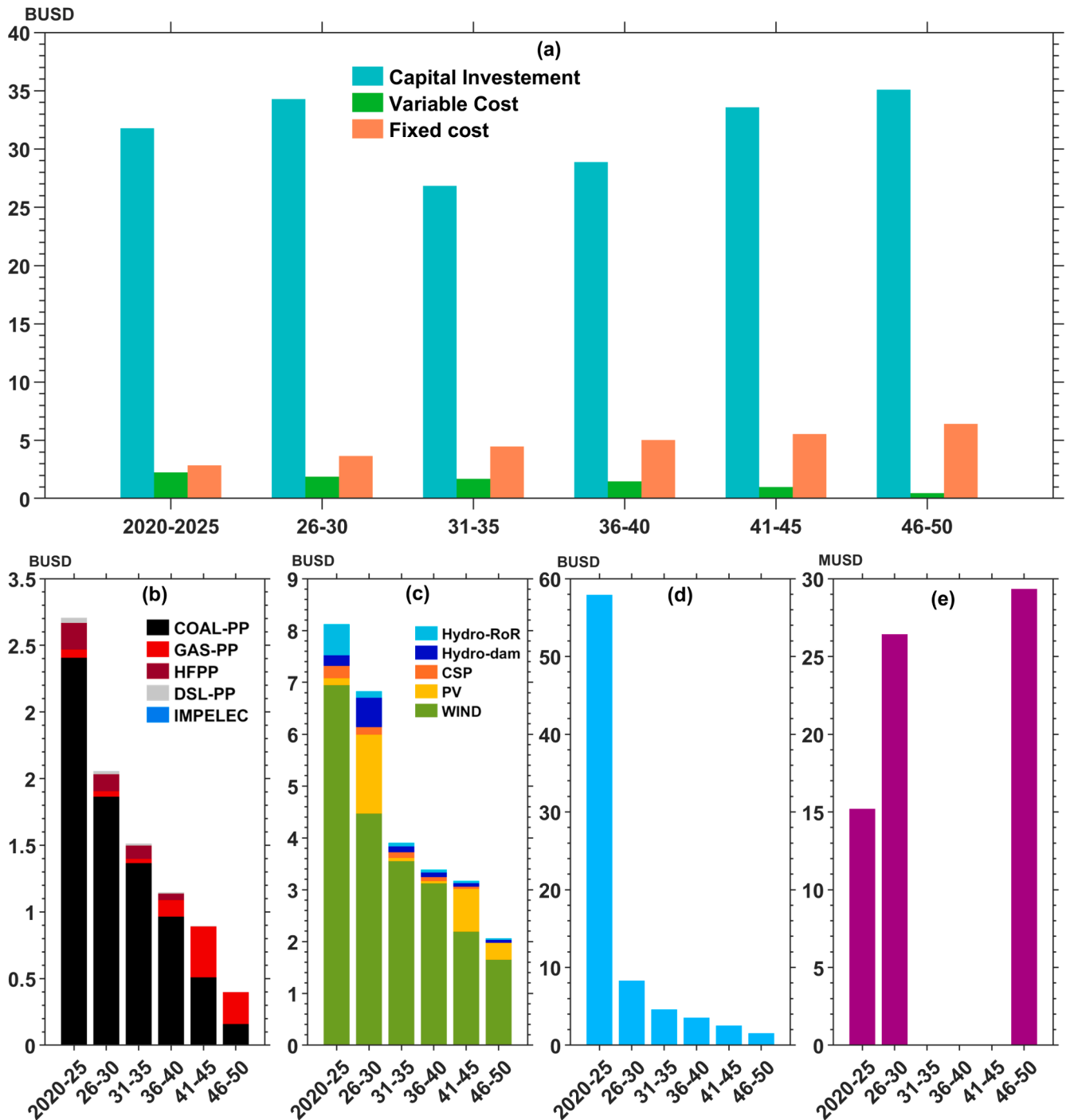


Fig. 4. Total system costs in the "BAU" scenario: (a) Investment and operating costs. (b) Breakdown of conventional technology costs. (c) Breakdown of renewable technology costs. (d) Costs related to grid development. (e) Costs related to storage capacity.

on the BAU scenario, CO₂ emissions are expected to decrease from 16.75 million metric tons in 2020 to 2.41 million metric tons in 2050. The other scenarios predict emissions to be even lower. Achieving this goal requires the implementation of RE projects and the decommissioning of coal-fired power plants.

4.3. Installed capacity composition

According to the RNES scenario, electricity demand is expected to decrease by 5 % compared to the BAU scenario from 2030 onwards. The RNES scenario assumes that government projects are completed on time

and without delays. Additionally, this scenario requires a specific breakdown of the 52 % share of RE capacity targeted for 2030, with wind and solar power each accounting for 20 % (including 16 % for PVs and 4 % for CSP) and hydropower for 12 %. These assumptions have resulted in significant differences in the projected electricity generation capacity, as illustrated in Fig. 5. Despite having 1.5 GW less generation capacity than the BAU scenario by 2050, the RNES scenario shows a similar trend, with renewables accounting for 73 % of the portfolio in 2050. Wind power remains the dominant generation technology in both scenarios, followed by solar thermal, PVs, CSP, hydro, and fossil fuels. The RNES scenario also reduces the capacity developed in PHS

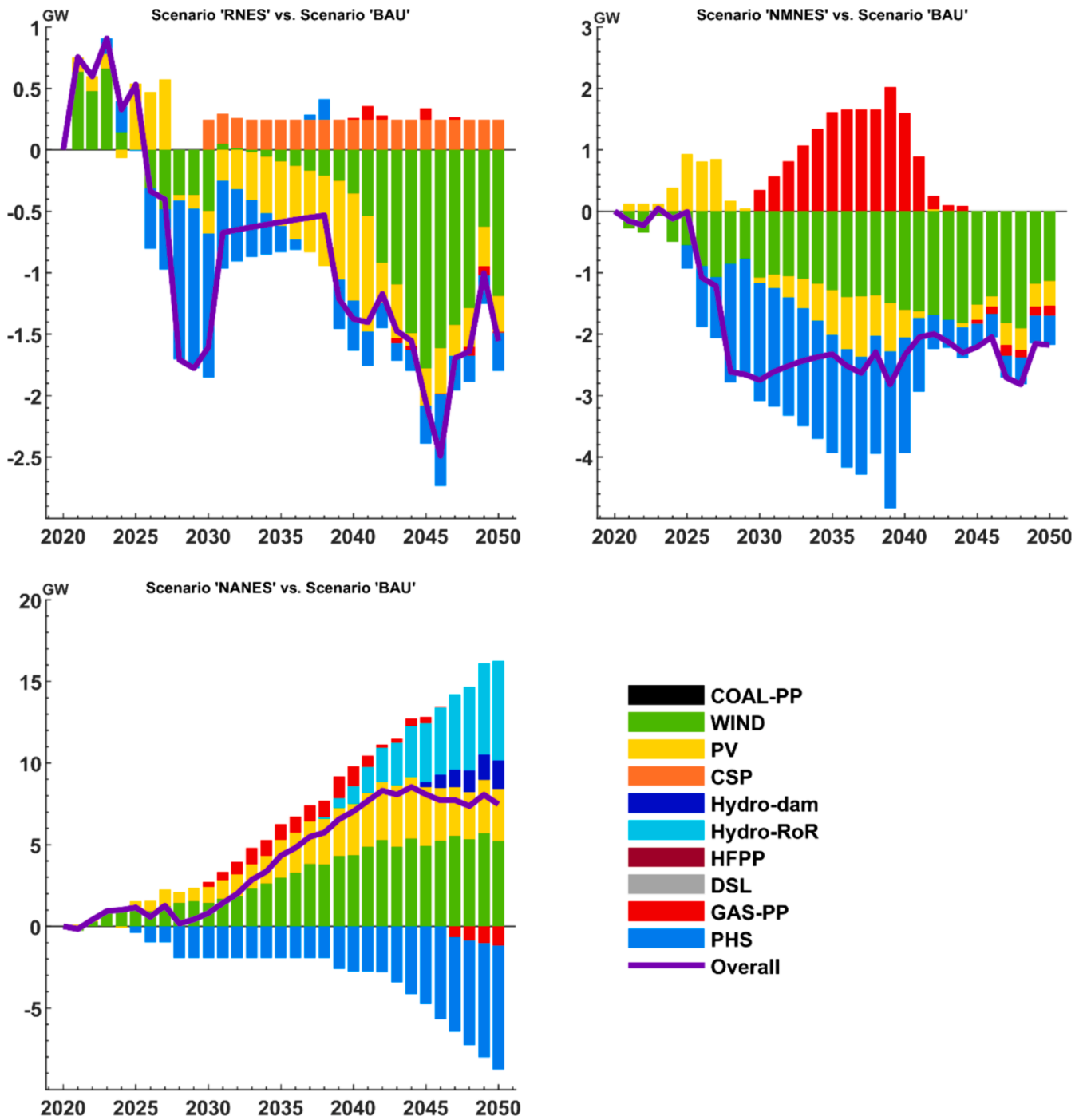


Fig. 5. Comparison of Production Capacities by Technology between the RNES, NMNES, and NANES Scenarios and the BAU Scenario. The bars show the difference in capacity between the two scenarios by Technology. The line shows the total difference each year.

technologies by 310 MW compared to the BAU scenario by 2050. Finally, the installed capacity of gas-fired power plants, although developed in different years, remains almost identical in both scenarios.

According to the information presented in Fig. 5, it appears that the NMNES scenario requires less investment in electricity generation capacity compared to the BAU scenario. In fact, by 2050, the NMNES scenario projects a 2.18 GW decrease in electricity generation capacity compared to the BAU scenario. While both scenarios achieve a similar adoption of renewable technologies (73 % by 2050), the installed renewable capacity is expected to be 1.54 GW lower in 2050 in the NMNES scenario. This reduction in capacity can mainly be attributed to the assumed 10 % decrease in demand starting in 2030, which is

achieved through EE measures. This allows the system to reach its renewable technology targets ahead of schedule, between 2028 and 2039, in the NMNES scenario.

An examination of energy investment trends under the NMNES scenario reveals significant discrepancies compared to the BAU scenario. Notably, wind power investments are lower in the NMNES scenario, with an average difference of 1.14 GW. This indicates a consistently higher investment in wind power within the BAU scenario throughout the modeling period. Conversely, solar PV investments display a distinct pattern, with a rise observed in the NMNES scenario from 2020 to 2029, surpassing the BAU scenario by an average of 350 MW. However, this trend shifts from 2030 onwards, resulting in a drop in solar PV

investments in the NMNES scenario by an average of 390 MW compared to the BAU scenario. Investments in gas-fired power generation also exhibit divergences. Between 2030 and 2044, NMNES investments outstrip those in the BAU scenario by an average of 1.04 GW. Nonetheless, investments in the NMNES scenario decreased after this period, averaging 130 MW lower than the BAU scenario from 2045 onwards. Furthermore, the use of PHS technology remains persistently lower in the NMNES scenario than in the BAU scenario throughout the modeling period, with an average deficit of 1.28 GW. These investment patterns offer valuable insights into the potential paths of energy infrastructure development in both scenarios.

When comparing the NANES and BAU scenarios, there are notable differences in electricity generation capacity and RE investments, as illustrated in Fig. 5. The NANES scenario proposes EE measures that would result in a 15 % reduction in electricity demand by 2030. Furthermore, by 2050, the NANES scenario is projected to have 12.92 GW more electricity generation capacity than the BAU scenario, with RE investment surpassing the BAU scenario by 16.26 GW. The increase in investment varies by technology, with wind and solar PVs demonstrating the most significant growth at 3 GW and 1.9 GW, respectively, throughout the modeling period. Although gas-fired power generation capacity in the NANES scenario increases by an average of 720 MW more than in the BAU scenario between 2030 and 2045, this trend is reversed from 2046 onwards, leading to a 930 MW drop in investment by 2050. Finally, while investment in hydropower from dams and rivers is, on average, 1.14 GW and 3 GW higher in the NANES scenario than in the BAU scenario, investment in PHS is 3 GW lower.

4.4. Electricity supply

According to Fig. 6, the RNES scenario is expected to generate slightly less electricity than the BAU scenario, with a difference of approximately 1.27 TWh, resulting in an estimated supply of 85.41 TWh in 2050. This reduction is thanks to the implementation of EE measures, leading to a decline in electricity demand. The electricity supply mix remains mostly the same in both scenarios, with RE making up 90 % of the supply by 2050. However, the RNES scenario has a higher maximum penetration of solar CSP technologies, reaching 6 % in 2030, compared to 4 % in the BAU scenario. In 2020, coal was the primary energy source in both scenarios, accounting for 61 % of the supply. However, from 2030 onwards, wind power will dominate, supplying 70 % in the BAU scenario and 73 % in the RNES scenario by 2050. The second most significant energy source varies between the scenarios, with wind power supplying 27 % of total electricity in 2020 and coal becoming the second most important source by 2035, contributing 25 % in the RNES scenario and 33 % in the BAU scenario. By 2050, solar PVs are expected to be the second most significant source, contributing 17 % in the RNES scenario and 15 % in the BAU scenario.

Furthermore, in both scenarios, diesel and fuel oil technologies will only be utilized as a last resort for electricity generation during emergencies. Under the RNES scenario, solar CSP technology is projected to contribute less than 1 % of the total electricity, while it will be nearly non-existent in the BAU scenario. In contrast, hydropower is expected to provide 2 % of the total electricity in the RNES scenario, compared to 1.5 % in the BAU scenario. In terms of electricity imports, their contribution to production is anticipated decreasing from 0.33 % in 2020 to

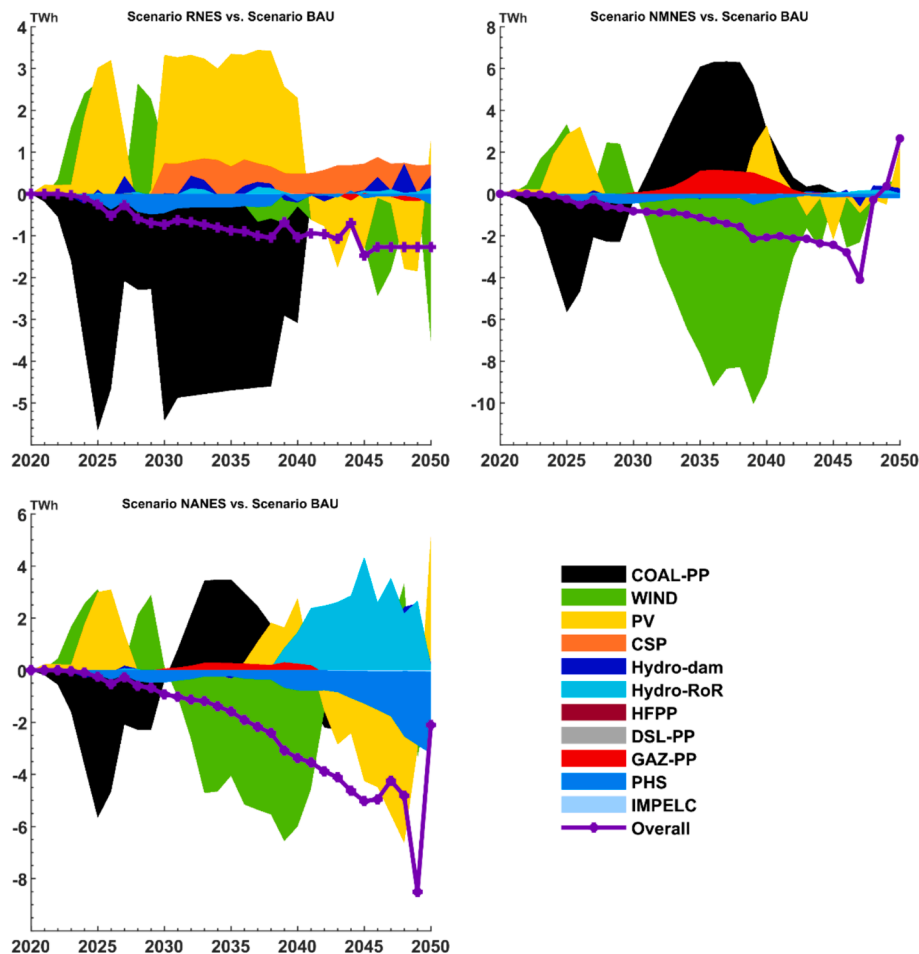


Fig. 6. This chart compares the energy produced by different technologies in different scenarios. The areas show the difference between the energy produced by the alternative scenario and that produced by the BAU scenario for the same technology. The line shows the difference in total production for each year.

0.08 % in 2050 in both scenarios. Gas-fired power plants will make up 3 % of the electricity generated in 2050 in both scenarios, while PHS technologies will account for 5 % of the electricity generated in 2050 in both scenarios. These projected percentages are primarily intended to ensure a sufficient reserve margin.

According to data presented in Fig. 6, in 2050, the NMNES scenario is projected to generate an additional 2.65 TWh of electricity compared to the BAU scenario, resulting in a total of 89 TWh. RE sources are expected to play a bigger role in the NMNES scenario, contributing approximately 3 TWh more in 2050. This will boost the share of RE technologies from 39 % in 2020 to 91 % in 2050, surpassing the 90 % estimated in the BAU scenario. Despite the decrease in electricity demand due to EE measures, the NMNES scenario aims for RE capacity targets of 67 % in 2050. This translates to a slight increase in electricity supply from renewable resources, particularly in solar PVs, averaging 420 GWh between 2020 and 2050. On the other hand, wind power generation is expected to decline by an average of 2.3 TWh in the NMNES scenario compared to the BAU scenario over the same period, while hydropower generation remains constant in both scenarios.

In the NMNES scenario, the integration of fossil fuel technologies is separated into two distinct phases. Between 2020 and 2030, the NMNES scenario demonstrates an average reduction of 2.12 TWh in coal usage in comparison to the BAU scenario. However, from 2031 to 2045, the trend flips, with the NMNES scenario exhibiting an average positive difference of 3.26 TWh compared to the BAU scenario. It is important to note that while the NMNES scenario permits the development of gas-fired power plants earlier than the BAU scenario (2030 vs. 2040), the electricity provided by this technology in the NMNES scenario surpasses the BAU scenario by only 570 GWh on average between 2030 and 2040. Additionally, diesel and oil-fired systems are solely employed to generate electricity during periods of peak demand.

According to the data in Fig. 6, in the NANES scenario, electricity supply is projected to reach 84.26 TWh by 2050, which is approximately 2.1 TWh less than the BAU scenario for the same year. This reduction is attributed to the EE measures implemented within the NANES framework, leading to a 15 % decrease in electricity demand starting in 2030. In addition, the NANES scenario predicts a higher share of RE sources, with 98 % of demand being met by RE sources in 2050, compared to 90 % in the BAU scenario. However, there are slight differences in the electricity sources between the two scenarios. The supply of electricity from solar PVs is expected to decrease slightly in the NANES scenario compared to the BAU scenario. Conversely, wind technology is expected to decrease by an average of 1.34 TWh between 2020 and 2050 in the NANES scenario compared to the BAU scenario. However, the increased use of hydropower technologies, with an average of 550 GWh over the same period, is the main reason for the higher penetration of renewables

in the NANES scenario.

In the NANES plan, the utilization of fossil fuel technologies is split into three phases. Between 2020 and 2030, coal usage decreased by an average of 2.13 TWh compared to the BAU scenario, while other fossil fuels remained steady. However, between 2031 and 2039, the trend reverses, and the NANES plan experiences an average increase of 2.46 TWh in comparison to the BAU scenario. This increase in fossil fuel technologies also applies to gas-fired power plants, which encounter an average increase of 220 GWh in the NANES scenario compared to the BAU scenario. From 2040, the downward trend continues, with an average decrease of 1.67 TWh for coal and 790 GWh for natural gas.

4.5. Use of storage capacity

Between 2020 and 2038, the average energy required to charge hydroelectric storage was 39 % lower in the RNES scenario compared to the BAU scenario (Fig. 7). However, from 2039 onward, this difference narrowed to just 1 %. Furthermore, the average energy derived from hydroelectric storage was 52 % higher in the BAU scenario, but this discrepancy also decreased to just 1 % from 2039 onward. These significant differences between the BAU and RNES scenarios can be attributed to project delays, particularly in wind power technology, in the BAU scenario. This delay has a substantial impact on the power system, requiring RE to be stored when not needed and used when demand is high. However, from 2039 onward, the two scenarios exhibit similar behavior in the power system.

The NMNES scenario indicates a 26 % reduction in energy consumption between 2020 and 2050 in comparison to the BAU scenario (Fig. 7). Additionally, the use of energy from hydraulic storage is, on average, 33 % lower in the NMNES scenario than in the BAU scenario. These differences in storage capacity for charging and discharging energy in the NMNES scenario stem from the rapid development of gas-fired power plants, providing reserve capacity and reducing reliance on pumped-storage technology, in contrast to the BAU scenario.

The NANES scenario predicts a 48 % reduction in the load energy of hydraulic storage capacities and a 49 % decrease in hydraulic storage energy consumption between 2020 and 2050, compared to the BAU scenario (Fig. 7). These changes are attributed to the significant increase in installed hydroelectric capacity, particularly run-of-river plants, which are known for their constant operation compared to the fluctuations of RE.

4.6. Total system cost comparative analysis

According to Fig. 8, the RNES scenario requires a capital expenditure of USD 3.8 billion less than the BAU scenario during the 2020–2050

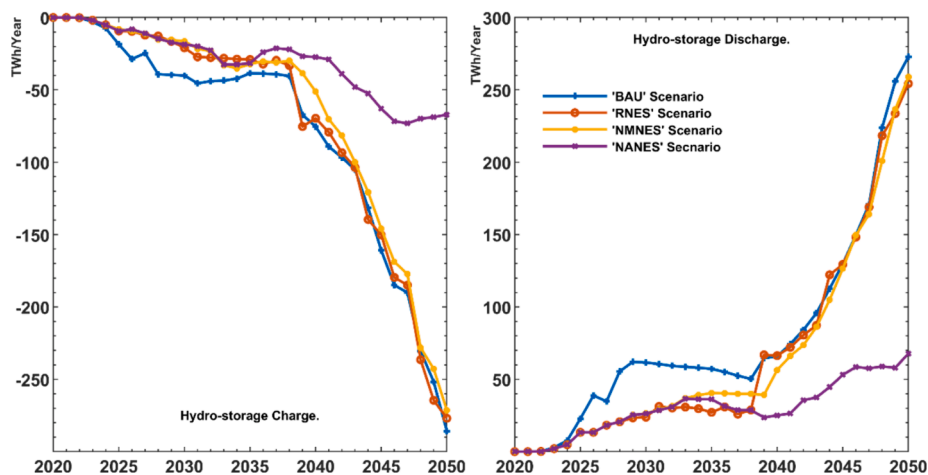


Fig. 7. The energy used for charging and discharging hydraulic storage in each scenario.

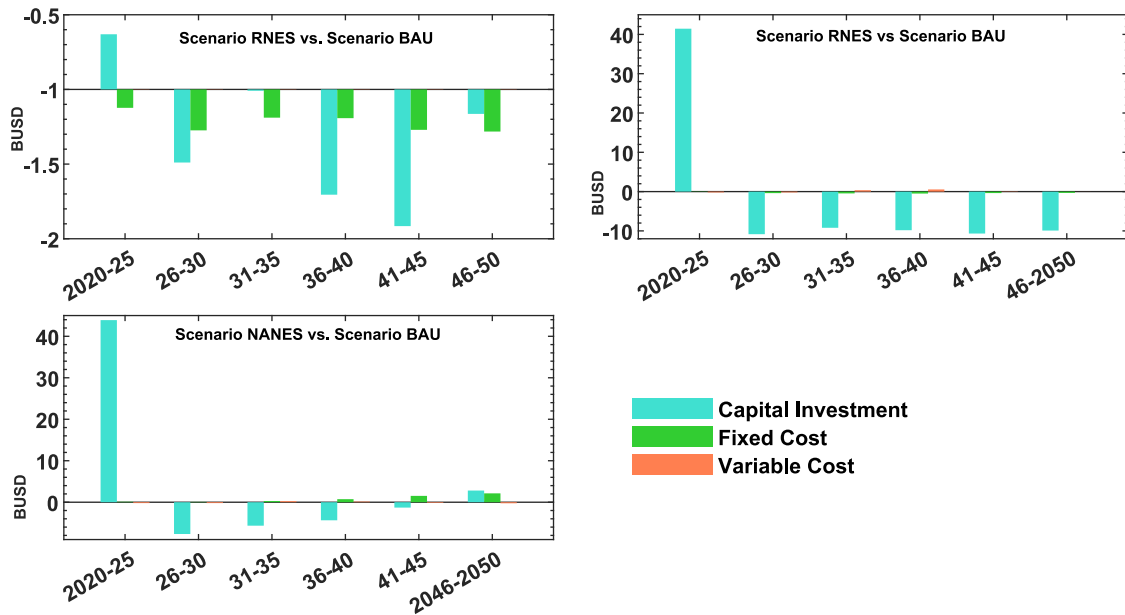


Fig. 8. This chart compares the costs of different scenarios. The bars show how much each scenario costs more or less than the BAU scenario. Negative values mean the alternative scenario costs are less than the BAU scenario.

period. Despite both scenarios achieving a RE penetration rate of 52 % in 2030, the RNES scenario recommends EE measures that result in a reduced installed capacity to meet demand. Consequently, the total system capacity is lower than that of the BAU scenario (refer to section 4.1). To achieve the energy transition objectives and fulfill demand between 2020 and 2050, the RNES scenario requires a budget of USD 45 billion for renewable technologies, USD 3.5 billion for conventional technologies (mainly gas-fired power plants), and USD 2.5 billion for PHS technologies. Additionally, an allocation of USD 136 billion is made for the expansion of T&D networks over the same period. The RNES scenario experiences a significant reduction in fixed costs, amounting to USD 2.6 billion, in contrast to the BAU scenario. However, variable costs remain the same between the two scenarios from 2020 to 2050. The decrease in fixed costs can be attributed to the limited deployment of wind, hydro, and PHS technologies in the RNES scenario relative to the BAU scenario.

In the more efficient NMNES scenario, the implementation of EE measures will result in a reduction of demand by 10 % as early as 2030. It will result in a total saving of USD 9 billion in investment costs compared to the BAU. The NMNES scenario sets higher renewable energy integration targets compared to the BAU scenario, with a goal of 67 % by 2050 versus 52 % by 2030 in the BAU scenario. Despite this ambitious target, the NMNES scenario demonstrates that the energy transition can be achieved at a lower cost (as illustrated in Fig. 8). In order to achieve the energy transition targets by 2050 in the NMNES scenario, an investment of USD 43 billion must be made in renewable technologies, USD 3.4 billion in gas-fired power plants, and USD 7.8 billion in PHS technologies between 2020 and 2050. Furthermore, an additional investment of USD 133 billion will be made in the development of T&D networks over the same period. Moreover, in the NMNES scenario, fixed costs are USD 2 billion lower than in the BAU scenario due to the reduced use of fossil and hydro technologies. In comparison, variable costs are USD 535 million higher between 2020 and 2050 due to the increased use of RE, resulting in greater demand for backup technologies such as gas.

According to Fig. 8, it is expected that the NANES scenario will require an additional USD 27.7 billion investment compared to the BAU scenario from 2020 to 2050. While the NANES scenario shows the highest demand reduction rate, it also sets more ambitious energy transition goals, resulting in higher investment costs of up to USD 218

billion during the same period. The need for greater renewable electricity generation capacity in the NANES scenario leads to higher overall costs than in the BAU scenario. To meet energy transition targets from 2020 to 2050, a total investment of USD 86 billion for renewable technologies, USD 2.4 billion for gas-fired power plants, and USD 258 million for PHS technologies would be required. Additionally, extending the T&D network would require an investment of USD 130 billion during the same period. The NANES scenario has fixed costs of USD 4.8 billion more than the BAU scenario, while variable costs are USD 429 million less than the BAU scenario from 2020 to 2050. The increase in fixed costs is mainly due to the higher adoption of hydroelectric and solar PV technologies.

4.7. A comparative analysis of CO₂ emissions

The Moroccan government is committed to creating a modern electricity system that can meet future energy needs while reducing GHG emissions between 2020 and 2050. According to the BAU scenario, CO₂ emissions in Morocco are projected to decrease from 16.75 Mt CO₂ in 2020 to 2.41 Mt CO₂ in 2050, with a peak of 19.62 Mt CO₂ in 2026. Suppose RE projects are implemented by 2030, and the decommissioning date for coal-fired power plants is met. In that case, it is estimated that CO₂ emissions will not exceed 10 Mt CO₂ after 2043 under the BAU scenario. The other three scenarios, RNES, NMNES, and NANES, will lead to CO₂ emissions of 2.39 Mt, 2.35 Mt, and 0.29 Mt, respectively, in 2050, compared to 16.75 Mt in 2020. The NANES scenario has the lowest electricity demand of all four scenarios and a significant increase in RE sources after 2030, resulting in lower CO₂ emissions. Nevertheless, the CO₂ emission levels of the BAU, RNES, and NMNES scenarios are also low (Fig. 9). The sharp decline in CO₂ emissions observed in the NANES scenario can be attributed to the influence of emission penalties that encourage the adoption of cleaner technologies due to the additional financial burden imposed by the penalty on the system.

4.8. Model validation

In order to confirm the operational reliability of the OSeMOSYS-Morocco model, we verified the energy generation balance based on the power system diagram designed for Morocco (refer to Figure S1 in

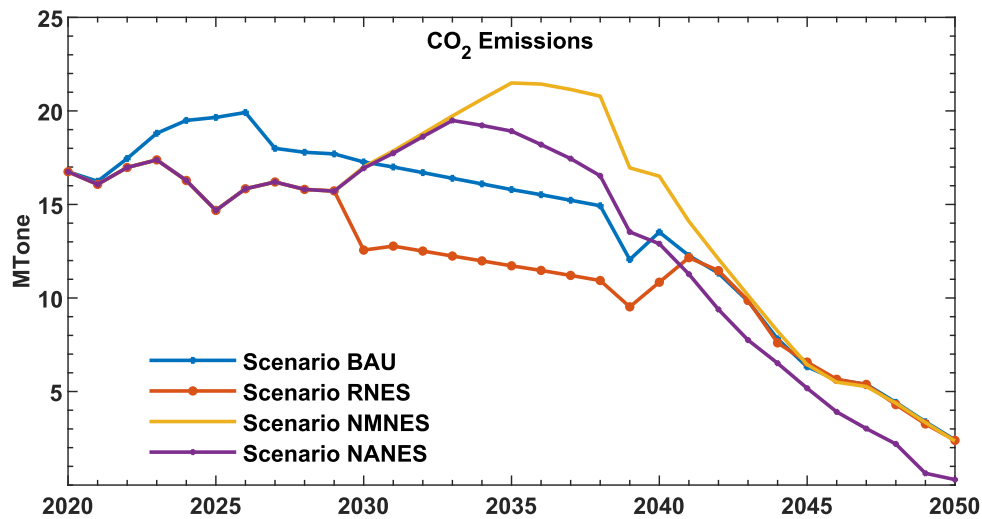


Fig. 9. Comparing CO₂ emissions from different scenarios.

the [supplementary document](#)). In 2020, under the BAU scenario, energy generation from various technologies amounted to 31.66 TWh, with 31.12 TWh transmitted via transmission lines and 30.28 TWh through distribution lines, meeting the total domestic demand [35]. T&D losses account for the decrease in energy from generation to final demand. This pattern is observed in all other scenarios and years, thereby confirming the reliability of the model. Moreover, these values are consistent with the ONEE reports for the years 2021 and 2022 [35].

5. Discussion and policy recommendations

Morocco is committed to establishing a sustainable and eco-friendly electricity supply system by leveraging its ample reserves of solar and wind energy. However, the unpredictability of these technologies and the limited availability of fossil fuels, especially natural gas, to support RE sources pose a significant challenge for the nation. To tackle these challenges, our research delves into four distinct scenarios to examine how diverse factors—such as growth in electricity demand, integration of RE, and scarcity of natural gas—could impact Morocco's energy transition. ([Supplementary materials](#) contain country development indicators.).

Morocco boasts an abundance of renewable resources that can be leveraged to modernize and improve its electricity supply system, ensuring efficiency and reliability. With the country's ubiquitous access to electricity and the lack of fossil fuels, effective energy system planning is vital. The energy sector has also been hindered by the disruption caused by the GME pipeline's natural gas supply interruption from Algeria [53]. Achieving sustainable energy development requires viable markets, appropriate energy policies, and political will [1]. To answer the original research question, this study identified electricity generation technologies that the government can invest in to provide affordable, cost-effective electricity while meeting energy strategy targets and future electricity demand up to 2050. Additionally, the study evaluated the environmental implications and electricity storage capabilities.

To achieve its energy strategy goals and meet future energy demands, the country must increase its renewable electricity generation capacity. However, it is important to consider that the power system's evolution may vary depending on the electricity demand and consumption levels in different scenarios. As detailed in section 3.3, if EE measures lead to a significant decrease in electricity demand from 2030 onwards and the integration rates of RE are moderate (RNES and NMNES scenarios), the need for generation capacity will decrease during the modeling period.

PHS technologies offer promising support for the widespread

integration of RE. By introducing gas-fired power plants earlier in the modeling period (in scenarios such as NMNES and NANES), the nation could effectively diversify its energy mix while requiring less overall generating capacity (as demonstrated in the NMNES scenario). Furthermore, this approach could provide a more cost-effective path for transitioning to a fully RE system (as seen in the NANES scenario) with higher RE targets. Ultimately, such an approach would maximize the use of the country's abundant renewable resources.

In contrast, delaying investments in gas-fired power plants until 2040 in both the BAU and RNES scenarios would result in greater adoption of PHS and wind technologies. This is due to their complementary nature, and wind power's higher CF compared to solar power. It is worth noting that the use of solar CSP technology is limited to scenarios where specific policies mandate it, such as in the RNES scenario, as it is not as economically viable as solar PV and wind power.

It is important to note that if the expansion of natural gas usage in the power system accelerates after 2030, there will be sufficient backup power available. Nevertheless, this will result in a rise in CO₂ emissions between 2030 and 2040 due to the combination of emissions from gas-fired and coal-fired power plants, with the latter not being phased out until after 2040. Besides, as coal-fired power plants near the end of their operational life, RE sources such as wind and solar power will play a more significant role in the electricity production, providing cleaner energy and reducing reliance on imports. It is also crucial to consider that all scenarios underscore the necessity to enhance T&D networks, which will be pivotal in the future.

Morocco is currently facing three major challenges as it navigates its energy transition. The country's hydroelectric resources are heavily reliant on rainfall, and their decline in recent years has resulted in most large dams not reaching their maximum filling capacity [66]. Despite the increasing integration of RE, there is a growing need for investment. Although Morocco has shown promising potential and attracted some international investment for RE projects, securing financing for large-scale projects remains challenging [1]. The transition to RE sources has social implications, including changes in employment patterns, land use, and energy costs. These changes can bring both benefits and difficulties during the transition.

Our research aimed to explore the advantages of using a novel renewable constraint based on installed capacity rather than renewable generation when modeling energy systems in developing nations. This approach facilitates better planning and management of the integration of RE into total installed capacity, leading to superior outcomes. Our findings indicate that the 52 % target for installed RE capacity can be achieved even with a three-year delay for solar and a two-year delay for

wind. According to the BAU scenario, the integration rate of RE should reach 57 % in 2030. The RNES scenario assumes that planned solar and wind projects are completed on time and requires a specific allocation of the 52 % target for total installed capacity in 2030. The result is a decrease in total installed capacity and an improvement in the integration rate of RE, which is expected to reach 59 % in 2030. Moreover, our extended modeling framework has enabled us to evaluate the feasibility of a new energy strategy up to 2050, including the NMNES and NANES scenarios.

The use of hydroelectric storage capacity can differ depending on various factors, such as the implementation of energy-efficient measures that can reduce the required installed capacity to meet energy demand. This effect is noticeable even when RE integration rates exceed 73 % in RNES and NMNES scenarios by the year 2050. However, in the NANES scenario, where RE integration rates increase to 92 % by 2050, the need for hydroelectric storage capacity decreases due to the expanded installation of river hydroelectric capacity.

To meet its energy goals, Morocco must make substantial investments in its electricity infrastructure. The government's plans for the future require capital expenditures ranging from USD 181 billion (under the NMNES scenario) to USD 218 billion (under the NANES scenario) from 2020 to 2050. On the other hand, estimated operating and maintenance expenses for the same period range from USD 34 billion (under the RNES scenario) to USD 41 billion (under the NANES scenario), with most of these costs being fixed expenses.

Morocco has committed to reducing its dependence on imported fossil fuels and transitioning to clean energy while maintaining low GHG emissions between 2020 and 2050. Under the BAU scenario, CO₂ emissions are expected to decrease annually from 16 Mt in 2020 to 2.41 Mt in 2050. Meanwhile, the RNES, NMNES, and NANES scenarios predict annual emissions of 2.39 Mt, 2.35 Mt, and 0.29 Mt, respectively, by 2050. However, cumulative emissions between 2020 and 2050 are estimated to be 421 Mt for the BAU scenario, 358 Mt for the RNES scenario, 446 Mt for the NMNES scenario, and 401 Mt for the NANES scenario. On average, this translates to 13.6 Mt for the BAU scenario, 11.5 Mt for the RNES scenario, 14.4 Mt for the NMNES scenario, and 12.95 Mt for the NANES scenario.

Our analysis indicates that Morocco should prioritize wind power in its upcoming RE projects to accelerate the transition towards sustainability. While this may entail higher costs, effective cost management of the system is feasible. The NANES scenario provides a viable roadmap towards achieving carbon neutrality, complete integration of renewable sources, and improved energy autonomy. Nonetheless, the success of this approach will hinge on the deployment of complementary technologies such as gas-fired power plants or PHS systems. It may also be necessary to explore alternative strategies, such as marine PHS stations or large-scale electricity storage, given the constraints on gas imports and gas-fired power plants.

The research employed a unique linear modeling approach using OSeMOSYS that differs from the existing literature [44,67–69]. This approach takes into account the specific characteristics of the Moroccan energy and power system, resulting in a more precise representation. Our temporal model also incorporates the variability of RE sources. Furthermore, we have included data on resource availability, technology maturity, government plans, construction lead times, and labor costs in Morocco. Our electricity demand projections are based on a comprehensive analysis of the sector, which evaluates factors such as the potential for cost reduction through technological advancements, the challenges presented by high system losses, and the energy savings expected from future policies. Additionally, we estimated the capacity and lifespan of power plants based on current government plans.

As a result of the diverse range of approaches and scenarios, comparing the results of this research with others is a challenging task. However, our findings are consistent with previous research on the energy transition in Morocco, thereby supporting the feasibility and cost-effectiveness of transitioning from fossil fuels to renewables.

Numerous studies, such as those conducted by Benzohra et al. and Schinko et al., have reached similar conclusions [25,70]. Moreover, Bouabid et al. also suggest starting the transition with wind power and subsequently incorporating solar PVs [71]. Additionally, Slimani et al. emphasize the importance of natural gas as a transitional fossil fuel between coal and renewables [72].

6. Conclusion and perspectives

This study proposed a long-term energy modeling approach for Morocco, addressing the challenge of transitioning from reliance on imported fossil fuels to a sustainable future dominated by RE sources. This approach has provided a solid analytical framework to help the country's policymakers make informed decisions to achieve near 100 % RE integration by 2050. An enhanced OSeMOSYS framework was utilized, tailored to Morocco's needs as a developing nation with vast RE potential but limited fossil fuels. Fine temporal resolution was integrated to capture RE intermittency. New functional sub-blocks were incorporated to enhance the model's flexibility and precision for planning a low-carbon energy future.

Morocco could transition to a RE-based electricity system with a 92 % integration rate by 2050 for an additional \$32 billion total cost. Achieving this requires adopting the ambitious NANES scenario, which includes EE measures to reduce energy demand by 15 % between 2030 and 2050 compared to baseline forecasts. This scenario would also result in the lowest emissions rate of 0.29 MtCO₂e by 2050, paving the way for complete decarbonization. Furthermore, this policy shift could transform Morocco from a fossil fuel importer to the region's leading green energy exporter.

The principal conclusions that address the research questions are presented below:

1. Renewable energies are set to rise from 33 % of the energy mix in 2020 to 57 % in 2030, according to the BAU scenario, with significant contributions from wind and solar PV technologies. This will enable current RE targets to be met, but progress beyond 2030 will be limited.
2. In all scenarios, the most cost-effective energy supply mix ensuring reliability and reduced GHG emissions includes predominantly wind and solar power, supported by hydro technologies, pumped-storage stations, and gas-fired plants when available. Solar CSP technology, while not optimal, may be developed if required by policy, as in the RNES scenario.
3. The comparison between the RNES and BAU scenarios revealed the impact of delayed investment in RE projects. Both scenarios forecast a 73 % RE share by 2050, but the BAU scenario shows higher generation capacity and investment costs, including greater investment in pumped storage technologies. Between 2020 and 2038, the BAU scenario required 39 % more energy for hydroelectric storage and an additional \$3.8 billion in capital expenditure due to higher fixed costs from increased deployment of wind, hydro, and PHS technologies. This highlights the need for proactive catch-up plans to avoid costly delays in RE projects.
4. By 2050, implementing a strategy targeting 100 % RE would enable Morocco to achieve 92 % integration at a cost of \$218 billion, and just \$32 billion more than the total cost of a BAU scenario. Given Morocco's capacity to attract foreign investment, these funds could be raised [73]. Morocco's advancements in solar PV and wind power could reduce costs through industrial integration [35]. Environmentally, this strategy would yield the lowest emissions rate of 0.29 MtCO₂e by 2050, paving the way for complete decarbonization.
5. The study emphasized the necessity of a diversified energy mix and robust EE plans to ensure a rapid and feasible transition of Morocco's energy system. Implementing EE measures could significantly lower future energy demand, reducing the need for extensive new generation capacity. Moreover, investing in backup technologies, such as

pumped-storage hydroelectric plants, is crucial for maintaining grid reliability in a RE-dominated system.

This research contributed with a robust, tailored version of the OSeMOSYS model, providing a detailed framework for energy transition strategies in developing countries. The practical implications are profound, offering policymakers a clear pathway to sustainable energy, emphasizing the need for strategic investments in renewables and backup technologies like pumped-storage hydroelectric plants to ensure grid reliability. The model's flexibility allows for scenario analysis, aiding in the setting of realistic RE targets and optimizing resource allocation. Future applications of this model can extend to other developing nations facing similar challenges, fostering global progress toward sustainable energy goals. This study's value lies in its comprehensive approach, combining technical precision with policy relevance, thus serving as an essential tool for driving forward the RE agenda in developing contexts.

CRedit authorship contribution statement

Jabrane Slimani: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Abdeslam Kadrani:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology. **Imad El Harraki:** Writing – review & editing, Validation, Supervision, Software, Resources, Methodology. **El hadj Ezzahid:** Writing – review & editing, Validation, Supervision, Software, Resources, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.enconman.2024.118857>.

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